Subject Code	Math 2	Intermediate Algebra
Module Code	11.0	Polynomial Equations
Lesson Code	11.1.2	Definition and Solutions of Polynomial Equations (Part 2)
Time Frame		30 Minutes

IGNITE (PART 2)

TA: 30 Minutes* ATA**:

We now proceed to solving polynomial equations. We begin with the statement of the zero-product property.

Zero-Product Property

Given real numbers a and b, the product of a and b is zero when a is zero or b is zero (Charles & Pearson Inc., 2015, p. 286).

For instance, if we have: $2 \cdot b = 0$, we know that $b = 0$. Or, if we have $a \cdot 5 = 0$, then we know that $a = 0$. If we have $a \cdot b = 0$, then we know that $a = 0$ or $b = 0$. If we have the product of two terms with variable x, $(x+2)(x-1) = 0$, then $x + 2 = 0$ or $x - 1 = 0$.

The zero-product property is utilized when solving polynomial equations by factoring. Consider the following examples.

Example 1: Solving polynomial equations by factoring

Solve the following polynomial equation using factoring: $x^2 + 5x = -6$.

1. Write the polynomial equation in general form:

$$x^2 + 5x + 6 = 0$$

2. Factor the polynomial:

$$(x+3)(x+2) = 0$$

3. **Apply the zero-product property** with $a = (x+3)$ and $b = (x+2)$.

   Note: Since we are multiplying two factors here, $(x+3)$ and $(x+2)$ whose product is zero, either one or both of them are zero. Thus, the task of finding the solution becomes the task of finding what values of x make $x+3$ and $x+2$ equal to zero.

$$x + 3 = 0 \qquad \text{or} \qquad x + 2 = 0$$

4. Solve for each equation:

$$x = -3 \qquad \text{or} \qquad x = -2$$

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5. Check each solution. Checking your solution is important to ensure that you obtained the correct answer. If your checking results to a false statement, then either you have obtained an incorrect solution or you have made a mistake in your checking process. In any case, it is imperative that you check the solutions you obtain every time.

   For $x = -3$

$$x^2 + 5x = -6$$
$$(-3)^2 + 5(-3) \stackrel{?}{=} -6$$
$$9 - 15 \stackrel{?}{=} -6$$
$$-6 \stackrel{\surd}{=} -6 \text{ True}$$

   For $x = -2$

$$x^2 + 5x = -6$$
$$(-2)^2 + 5(-2) \stackrel{?}{=} -6$$
$$4 - 10 \stackrel{?}{=} -6$$
$$-6 \stackrel{\surd}{=} -6 \text{ True}$$

   The solutions are $x = -3, -2$.

This example also provides us with the general principles on how to solve polynomial equations by factoring (Charles & Pearson Inc., 2015, p. 286):

1. Write the polynomial equation in general form.
2. Factor the polynomial.
3. Apply the zero-product property.
4. Solve for each equation.
5. Check each solution.

We apply these principles in the examples that follow.

Example 2: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $2x^2 = 4x$.

1. Write the polynomial equation in general form:

$$2x^2 - 4x = 0$$

2. Factor the polynomial:

$$(2)(x)(x - 2) = 0$$

3. **Apply the zero-product property** with $a = 2$, $b = x$, and $c = (x - 2)$. Note here that the zero-product property can be extended to three or more terms. Observe however, that $a = 2$ will never be zero. So, we only consider the other two factors.

$$x = 0 \qquad \text{or} \qquad x - 2 = 0$$

4. Solve for each equation:

$$x = 0 \qquad \text{or} \qquad x = 2$$

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5. Check each solution.

   For $x = 0$

$$2x^2 = 4x$$
$$2(0)^2 \stackrel{?}{=} 4(0)$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

   For $x = 2$

$$2x^2 = 4x$$
$$2(2)^2 \stackrel{?}{=} 4(2)$$
$$8 \stackrel{\surd}{=} 8 \text{ True}$$

   The solutions are $x = 0, 2$.

Example 3: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $x^4 - 81 = 0$.

1. Write the polynomial equation in general form:

$$x^4 - 81 = 0$$

2. Factor the polynomial:

$$(x^2 - 9)(x^2 + 9) = (x - 3)(x + 3)(x^2 + 9) = 0$$

3. **Apply the zero-product property** with $a = (x - 3)$, $b = (x + 3)$, and $c = (x^2 + 9)$. Note here that the factor $x^2 + 9$ is no longer factorable. It is often called an irreducible quadratic factor because it cannot be reduced into factors that only contain real numbers (UT Austin Department of Mathematics, n.d.).

$$x - 3 = 0 \qquad \text{or} \qquad x + 3 = 0 \qquad \text{or} \qquad x^2 + 9 = 0$$

4. Solve for each equation. Note that for the irreducible quadratic factor, we used the square root property to solve for the value of x.

$$x = 3 \qquad \text{or} \qquad x = -3 \qquad \text{or} \qquad x = \sqrt{-9} = \pm 3i$$

5. Check each solution.

   For $x = 3$

$$x^4 - 81 = 0$$
$$(3)^4 - 81 \stackrel{?}{=} 0$$
$$81 - 81 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

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For $x = -3$

$$x^4 - 81 = 0$$
$$(-3)^4 - 81 \stackrel{?}{=} 0$$
$$81 - 81 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

For $x = 3i$

$$x^4 - 81 = 0$$
$$(3i)^4 - 81 \stackrel{?}{=} 0$$
$$81 - 81 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

For $x = -3i$

$$x^4 - 81 = 0$$
$$(-3i)^4 - 81 \stackrel{?}{=} 0$$
$$81 - 81 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

The solutions are $x = 3, -3, 3i, -3i$.

So far, we have seen that the solutions of the polynomial equations are distinct. But it is also possible that the polynomial equation can have the same solution twice or even thrice as we shall see in the following example.

Example 4: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $x^3 + x^2 - x - 1 = 0$.

1. Write the polynomial equation in general form:

$$x^3 + x^2 - x - 1 = 0$$

2. Factor the polynomial. In this case, we will **factor by grouping**. We group x^3 and $-x$ together as well as x^2 and -1 together because as we shall see, we will get a common factor of $x^2 - 1$.

$(x^3 - x) + (x^2 - 1) = 0$	*Factor by grouping*
$x(x^2 - 1) + (x^2 - 1) = 0$	*Take out common factor* $x^2 - 1$
$(x + 1)(x^2 - 1) = 0$	*Factor* $x^2 - 1$ *using the sum and difference of two squares rule.*
$(x + 1)(x + 1)(x - 1) = 0$	

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3. **Apply the zero-product property** with $a = (x + 1)$, $b = (x + 1)$, and $c = (x - 1)$.

$$x + 1 = 0 \qquad \text{or} \qquad x + 1 = 0 \qquad \text{or} \qquad x - 1 = 0$$

4. Solve for each equation:

$$x = -1 \qquad \text{or} \qquad x = -1 \qquad \text{or} \qquad x = 1$$

5. Check each solution.

For $x = -1$

$$x^3 - x + x^2 - 1 = 0$$
$$(-1)^3 - (-1) + (-1)^2 - 1 \stackrel{?}{=} 0$$
$$-1 + 1 + 1 - 1 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

For $x = 1$

$$x^3 - x + x^2 - 1 = 0$$
$$(1)^3 - (1) + (1)^2 - 1 \stackrel{?}{=} 0$$
$$1 - 1 + 1 - 1 \stackrel{?}{=} 0$$
$$0 \stackrel{\surd}{=} 0 \text{ True}$$

Notice that $x = -1$ is a double solution of the equation $x^3 - x + x^2 - 1 = 0$ because it was obtained from two repeated factors (Charles & Pearson Inc., 2015, p. 287). In general, solutions obtained from repeated factors are called **multiple solutions** (of which double solution is a subcategory; for instance, three repeated factors give rise to a triple solution) (Charles & Pearson Inc., 2015, p. 287). In reporting multiple solutions, we include the **multiplicity** of the solution. From our earlier example, we have the following solutions: $x = 1, -1$ (**with multiplicity two**) because -1 appeared twice as a solution.

It is important to point out here that the number of solutions for a polynomial equation is equal to its degree. In our example above, we had a third degree polynomial, so we were expecting to have three (3) solutions as well, which we did. This is also why indicating the multiplicity is important. If we only reported the solutions as follows: $x = 1, -1$ (that is, without multiplicity), the number of solutions would only be two (2), which is wrong because our polynomial equation is degree 3. But by reporting the multiplicity, we get three (3) solutions, which is correct in terms of the degree of the polynomial.

Example 5: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $x^3 = 8$.

1. Write the polynomial equation in general form:

$$x^3 - 8 = 0$$

2. Factor the polynomial:

$$(x - 2)(x^2 + 2x + 4) = 0$$

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3. **Apply the zero-product property** with $a = (x - 2)$ and $b = (x^2 + 2x + 4)$.

$$x - 2 = 0 \qquad \text{or} \qquad x^2 + 2x + 4 = 0$$

4. Solve for each equation:

$$x = 2$$

or

$$x^2 + 2x + 4 = 0$$

Using quadratic formula, the lower equation can be solved as:

$$x = \frac{-2 \pm \sqrt{4 - 16}}{2}$$

$$x = \frac{-2 \pm \sqrt{-12}}{2}$$

$$x = \frac{-2 \pm 2i\sqrt{3}}{2}$$

$$x = -1 \pm i\sqrt{3}$$

Note: $x^2 + 2x + 4$ here is an irreducible quadratic equation or a prime polynomial. One way to know if a quadratic equation is irreducible or not is to solve for its discriminant. Recall that the discriminant is given by the expression: $b^2 - 4ac$. If $b^2 - 4ac > 0$ ***and*** *a perfect square or when $b^2 - 4ac = 0$, then the quadratic equation is factorable. In our case, $a = 1$, $b = 2$, and $c = 4$, the value of the discriminant is -12. So, we know that indeed this polynomial is not factorable since its discriminant is negative which means that the roots are imaginary numbers.*

5. The checking for this example will be left to you as a practice exercise.

Example 6: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $2x^4 - 2 = 3x^2$.

1. Write the polynomial equation in general form:

$$2x^4 - 3x^2 - 2 = 0$$

2. Factor the polynomial. This is an equation leading to quadratic forms. To see this, we let $u = x^2$.

$$2u^2 - 3u - 2 = 0$$
$$(2u + 1)(u - 2) = 0$$

3. **Apply the zero-product property** with $a = (2u + 1)$ and $b = (u - 2)$.

$$2u + 1 = 0 \qquad \text{or} \qquad u - 2 = 0$$

4. Solve for each equation:

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$$u = -\frac{1}{2} \qquad \text{or} \qquad u = 2$$

Going back to the variable x via $u = x^2$, we have:

$$x^2 = -\frac{1}{2} \qquad \text{or} \qquad x^2 = 2$$

Applying the square root property, we have:

$$x = \pm\sqrt{-\frac{1}{2}} = \pm i\frac{\sqrt{2}}{2} \qquad \text{or} \qquad x = \pm\sqrt{2}$$

5. The checking for this example will be left to you as a practice exercise.

Example 7: Solving polynomial equations by factoring

Solve the polynomial equation using factoring: $(x-1)^3 = -(x-1)^2$.

1. Here, it would be useful to let $u = x - 1$. Then, we rewrite the resulting polynomial equation in general form:

$$u^3 = -u^2$$

$$u^3 + u^2 = 0$$

2. Factor the polynomial.

$$(u^2)(u+1) = 0$$

3. **Apply the zero-product property** with $a = (u^2)$ and $b = (u+1)$.

$$u^2 = 0 \qquad \text{or} \qquad u + 1 = 0$$

4. Solve for each equation:

$$u = 0 \text{ (multiplicity of two)} \qquad \text{or} \qquad u = -1$$

Going back to the variable x via $u = x - 1$, we have:

$$x - 1 = 0 \text{ (multiplicity of two)} \qquad \text{or} \qquad x - 1 = -1$$

We then have:

$$x = 1 \text{ (multiplicity of two)} \qquad \text{or} \qquad x = 0$$

5. The checking for this example will be left to you as a practice exercise.

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As a reference, below is a table of special factors to aid you in determining the factors of some common polynomials.

TABLE OF SPECIAL FACTORS	
POLYNOMIAL	**FACTORED FORM**
$a^2x^2 + 2abx + b^2$	$(ax + b)^2$
$a^2x^2 - 2abx + b^2$	$(ax - b)^2$
$a^2x^2 - b^2$	$(ax - b)(ax + b)$
$a^3x^3 + b^3$	$(ax + b)(a^2x^2 - axb + b^2)$
$a^3x^3 - b^3$	$(ax - b)(a^2x^2 + axb + b^2)$
$a^4x^4 - b^4$	$(a^2x^2 + b^2)(ax + b)(ax - b)$

Note: a and b here are non-zero rational numbers.

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Prepared by: David Bryan C. Lao
Position: Special Science Teacher (SST) I
Campus: PSHS-Western Visayas Campus

Reviewed by: Zennifer L. Oberio
Position: Special Science Teacher (SST) IV
Campus: PSHS-Western Visayas Campus

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